



No. of Printed Pages: 02

Seat No: _____

[98]

SARDAR PATEL UNIVERSITY
M.Sc (IT) Semester-I Examination – November 2022
PS01CINT54: Relational Database Management System
Friday, 11th November 2022

Marks: 70

Time: 10.00am To 1.00pm

[08]

Q-1 Select an appropriate option for the following statement.

- 1 _____ is a systematic or organized collection of related information.
[a] Database [b] Table [c] Rows [d] Columns
- 2 _____ level is the highest level in the three-level architecture and closest to the user.
[a] Internal [b] Conceptual [c] External [d] None of given
- 3 Multi valued attributes are represented by _____ symbol.
[a] Ellipse [b] Double ellipse [c] Dashed ellipse [d] None of given
- 4 The _____ join produces table that merges each row of the first table with each row of the second table.
[a] cross [b] inner [c] outer [d] full
- 5 _____ returns always one value to a caller.
[a] Procedure [b] Trigger [c] Function [d] a & c both
- 6 In Oracle there are _____ numbered exception handlers.
[a] 10 to 15 [b] 15 to 20 [c] about 20000 [d] None of given
- 7 _____ cursor attribute returns number of records in a cursor.
[a] %Rowcount [b] %count [c] %Recordcount [d] None of given.
- 8 The _____ command is used to modify the records in a table.
[a] ALTER [b] UPDATE [c] MODIFY [d] None of given

[14]

Q-2 Answer the following questions in brief. (Attempt Any SEVEN)

- 1 Explain briefly inner SQL join.
- 2 Briefly discuss Data Integrity with its types.
- 3 Compare DBMS with RDBMS.
- 4 What is E-R Diagram? List its components.
- 5 What is Entity? Give suitable example.
- 6 List out at least 4 the most common data types available in PL/SQL.
- 7 Write the advantages of PL/SQL.
- 8 State the two differences between database Trigger & Stored procedure.
- 9 Write a SQL command to display the result of $500 * 200$.

Q-3 Do as directed.

- [A] Write a note on users associated with database system with their role.
[B] Write short note on concurrency control including its definition and techniques.

[06]
[06]

OR

- [B] Discuss different level of schema considering Three Tire Architecture.

[06]

Q-4 Do as directed.

- [A] Discuss Normalization with main purpose to adopt it. Also discuss any two forms of normalization of your choice with example.

[06]

- [B] What are CODD's rule? List and, briefly discuss any two rules from them.

OR

[06]

- [B] What do you mean by Relationship? Explain types of Relationship with suitable example.

[06]

Q-5 Do as directed.

- [A] Write SQL command for each of the following. [Any FOUR]
Assume that the table EMP having fields (ENO number(2,0), DNO number(1,0), ENAME varchar2(15), BASIC_PAY Number(8,0).

[06]

- Create table EMP having ENO as primary key.
- List out DNO and DNO wise total BASIC_PAY for all employees of EMP table.
- Remove all the records having ENO greater than 2 from a table EMP.
- Retrieve all employee information of table EMP in ascending order of ENAME.
- Display unique records of a table EMP.

- [B] Explain briefly the following Built-in function / SQL command with example:[Any FOUR]

[06]

- MOD()
- LENGTH()
- TRUNC()
- UPDATE
- DELETE

OR

- [B] Define constraints and write short note on types of constraints.

[06]

Q-6 Do as directed.

- [A] Explain briefly PL/SQL block structure with purposes of each section.
What do you mean by Function & procedure? Also write at least two differences between them.

[06]

- [B] Write short note on exception handling.

OR

[06]

- [B] Write a function named 'FSUB, which subtracts to given numbers say A and B. Also write PL/SQL code block to read two numbers N and M. Using above function FSUB subtract two numbers N and M and print the result.

[06]